Assignment: Decision Tree (DT) Implementation and Comparison

This assignment requires you to implement, tune, and compare two Decision Tree algorithms, CART (Gini) and ID3 (Entropy), within a single, fully automated machine learning workflow hosted on GitHub

1. GitHub & Automation (Mandatory)

1. Create a public GitHub repository
2. Upload your dataset (.csv) and your Colab Notebook (.ipynb)
3. Your Colab notebook must automatically load the dataset using a raw GitHub link
4. Manual file upload is **forbidden**

2. Pipeline Steps in Colab

Your notebook must include the following for the full workflow:

Step	Requirement
Data Preprocessing	Handle missing values. Encode categorical features
CART Implementation	Implement DecisionTreeClassifier using the Gini criterion. Tune max_depth and min_samples_split using Cross-Validation
ID3 Implementation	Implement DecisionTreeClassifier using the Entropy criterion. Tune max_depth and min_samples_split using Cross-Validation

3. Required Visualizations (2x1 Comparison)

Generate all required visualizations on the Test Set for both the optimized CART and ID3 models. Use a 2x1 matrix (side-by-side plots) for direct comparison where specified

Output	Type	Comparison Format (CART vs. ID3)
1. Decision Boundary	Plot visualization	2x1 Matrix
2. Confusion Matrix	Heatmap visualization	2x1 Matrix
3. ROC Curve	Plot visualization	2x1 Matrix
4. Evaluation Metrics	Bar Chart (Accuracy, Precision, Recall, F1, AUC)	Combined chart comparing both models
5. DT Structure	Visualization of optimized tree structure	Single image showing tree structure

4. Submission

- GitHub Link: The URL to your public repository containing both the dataset and the .ipynb file.
- Colab Link: The direct link to your notebook
- ***<u>I should be able to click the Colab link, run all cells, and have the dataset download and the model train automatically without manual intervention, with plots generated</u>***